

Snowflake Postgres :

Snowflake Postgres refers to **managing and using PostgreSQL databases alongside or integrated with Snowflake**, where:

- **Snowflake** is used for **analytics & data warehousing**
- **PostgreSQL** is used for **transactional (OLTP) workloads**

In simple words:

Snowflake Postgres = PostgreSQL for application data + Snowflake for analytics, working together in the same data ecosystem.

Feature	Snowflake	PostgreSQL
Type	Data Warehouse	Transactional Database
Best for	Analytics, reporting	App data, transactions
Data size	Very large (TBs–PBs)	Small to medium
Scaling	Auto & independent (storage/compute)	Manual / limited
Performance	Fast for large queries	Fast for small transactions
Maintenance	Fully managed	Needs admin work
Cost model	Pay for usage	Fixed / infra-based

Creating an instance :

- Creating a PostgreSQL instance means setting up a new PostgreSQL database server with its own CPU, memory, storage, and configuration.
- Think of a **Postgres instance** as a **separate database machine** where your data lives.
- One instance = one Postgres server and It can host **multiple databases**, It runs independently from other instances.

Example : A company has 3 environments.

Environment	PostgreSQL Instance
Development	dev-postgres
Testing	test-postgres
Production	prod-postgres

What happens when you create an instance?

- PostgreSQL version (e.g., 14, 15)
- CPU & memory size
- Storage size
- Network (VPC / private / public)
- Admin user & password
- Backup & security settings
- Each is a **separate Postgres instance**.

Create Instance → Configure → Start → Connect → Use Database

Connection:

Connecting in PostgreSQL means establishing a secure session between a user or application and a PostgreSQL database instance so they can read or write data.

Two Types in connection :

- Roles
- Connection Pooling

Roles :

- Roles define who can connect and what they are allowed to do in PostgreSQL.

A role = user + permissions

```
1 CREATE OR REPLACE USER report_user
2 PASSWORD = 'pwd'
3 DEFAULT_ROLE = report_user
4 MUST_CHANGE_PASSWORD = TRUE;
5
6 GRANT ROLE report_user TO USER report_user;
7
```

Results (just now)

Table Chart 1 row 144ms

id	status
1	Statement executed successfully.

Connection Pooling :

- Connection pooling means reusing a small number of database connections instead of opening a new connection for every request.

App → Connection Pool → PostgreSQL

- App asks pool for a connection
- Pool gives an existing connection
- After query → connection goes back to pool.

Monitoring in postgresQL :

Monitoring means tracking cost, performance, and activities of the database to keep it healthy, fast, and cost-efficient.

1. Evaluate Cost

- Tracks how much the database instance costs
- Shows usage of CPU, storage, and running time
- Understand where money is being spent

Why it matters:

- Avoids over-sizing databases
- Helps reduce unnecessary spend

Example : A database running 24/7 but used only 8 hours → monitoring shows wasted cost.

2. Insights :

- Gives deep visibility into:
 - Slow queries
 - High CPU or memory usage
 - Peak usage times

Why it matters:

- Helps tune queries
- Improves performance
- Prevents outages
- Understand performance and behavior

Example : Monitoring insights show one report query consuming 80% CPU every morning.

3. Logging :

- Stores logs for:
 - Errors
 - Connections
 - Query execution
 - Security events

Why it matters:

- Troubleshooting issues
- Auditing access
- Debugging failures
- Keep a record of everything happening

Example : A login failure occurs → logs show who tried, when, and from where.

Simple Hierarchy of Monitoring in PostgreSQL :

Monitoring

- | Evaluate Cost → Control spending
- | Insights → Improve performance
- | Logging → Troubleshoot & audit

Security and Networking

Security and networking ensure that the PostgreSQL database is accessed only by authorized users and only through secure network paths.

1. Networking

- Defines **how and from where** the database can be accessed
- Manages network paths between apps and the database
- Control where connections come from

What it controls:

- Public or private access
- Allowed IP addresses / VPCs
- Firewalls and security groups
- Port access (usually 5432)

Example : Only application servers inside the company network can connect to Postgres.

2. Tri-Secret Secure

- Advanced **encryption model** for data protection
- Protect data using multiple encryption keys
- Uses **three independent encryption keys**:
 1. Cloud provider key
 2. Snowflake-managed key
 3. Customer-managed key

Why it matters:

- Even if one key is compromised, data remains secure
- Meets strict compliance (banking, healthcare)

Example : Data is locked with three different keys—no single party can decrypt it alone.

Simple Hierarchy of Security and Networking:

Security and Networking

└ Networking → Control **access** paths

└ Tri-Secret Secure → Protect data **with** encryption